

Bias, Fairness, Governance and Sustainability Report

Project Title: Smart City Traffic Intelligence Capstone

Author: Chew De Sheng

Target Audience: City Government Smart-City Mobility Team

1. Bias and Fairness Report

1.1 Sampling and Coverage Limitations: The dataset used for model training relies entirely on continuous hourly traffic logs extracted from a single westbound corridor of the I-94 interstate highway near Minneapolis-St Paul. Because this data reflects a highly specific geographic environment, the predictive patterns developed cannot be generalized to alternative municipal layouts, different climates, or regions with distinct public transit infrastructure without introducing significant out-of-distribution error rates.

1.2 Proxy Label Risks: Due to the lack of access to an authenticated historical traffic accident database, a synthetic proxy label was engineered by intersecting heavy traffic volume splits (top two quartiles) with adverse weather indicators. The primary risk of this approach is the introduction of systemic operational blind spots. Real-world accidents are frequently driven by human variables entirely disconnected from weather—such as driver fatigue or distracted driving. Consequently, the model may underrepresent crash risks on clear, optimal business days.

1.3 Error Distribution Across Conditions: The target proxy label exhibits a severe historical class imbalance, with only 10.87% of observations satisfying the high-risk criteria. Because the machine learning models are exposed to an overwhelming volume of standard traffic conditions, the resulting errors are distributed unevenly. The system naturally achieves high performance during stable, predictable mid-day windows but suffers from reduced predictive precision during rare, sudden atmospheric shifts.

2. Governance and Sustainability Framework

2.1 Pre-Deployment Oversight: Before any predictive model is trusted to automate municipal infrastructure assets, a strict governance framework must be enforced. First, a human-in-the-loop operational policy must be implemented, ensuring that model outputs serve as decision support for human traffic controllers rather than executing autonomous updates. Second, a mandatory 30-day shadow deployment period must be completed to benchmark live API inference integrity against historical performance matrices.

2.2 Environmental and Compute Trade-offs: Advanced sequential neural network architectures like the Long Short-Term Memory (LSTM) network require significant computational resources and electricity during training loops compared to simple linear baselines. To mitigate this environmental footprint, this project successfully applied model quantization techniques. Downcasting model weights from standard float64 allocations to compact int8 formats reduced the physical file footprint by nearly half. This compression allows the model to execute local inference directly on low-power, solar-powered roadside smart cameras, cutting data-streaming bandwidth needs and minimizing regional cloud datacenter power consumption.